

Thomas Kopalidis

thomaskopalidis@yahoo.com

+30 697 849 7661

LinkedIn

GitHub

Personal Site

ABOUT ME

AI Engineer and Data Scientist with 4+ years building production ML/AI systems in regulated environments (healthcare, mobility, real estate), including engagements with PwC, CERTH and Tech Open House. Experienced in agentic AI systems, RAG architectures, and full-stack deployment. Familiar with EU AI Act and model governance principles.

Technical Skills

- **GenAI & LLMs:** LangGraph, CrewAI, Prompt Engineering, LLM Integration, RAG, Multi-agent systems.
- **AI Governance:** EU AI Act awareness
- **Programming & Data:** Python (PyTorch, Keras, Scikit-Learn, NumPy, Pandas), TensorFlow, SQL, R.
- **Cloud:** Azure (Databricks), AWS, Hugging Face Spaces
- **Visualization:** Power BI, Matplotlib, Seaborn
- **Tools:** Git, Docker, CI/CD, Flask, PostgreSQL, FastAPIs

WORK EXPERIENCE

AI Engineer (Project)

PwC Greece

GR (April 2026 - June 2026)

- Developed a production-ready Agentic RAG system for internal policy Q&A, using langGraph, FastAPI enabling multi-turn, context-aware responses over versioned policy documents (PDF), with step-by-step agent reasoning, dynamic tool selection, and evidence-grounded answers. Containerized the full stack with Docker; applied enterprise best practices including session memory management and scalable document ingestion. Version-controlled via Git.

Operational Data Analyst

Elysium Hospitality

Rhodes, GR (May 2025 - Nov. 2025)

- Designed executive dashboards (Excel/BI) to visualize Year over Year trends and demographics (e.g., top markets: Germany 27%, UK 15%), translating KPIs into revenue strategies.
- Managed daily financial reconciliation for 300+ rooms via Fidelio Opera PMS, analyzing peak traffic days (Sundays) to optimize staffing and resource allocation.

AI Engineer & Researcher

Centre for Research & Technology Hellas (CERTH ITI) | *EU Horizon Europe*

Thessaloniki, GR (Feb 2022 - Feb 2025)

EU Horizon Projects: UPPER · AI4HF · TeNDER · AutoTRUST · INTREPID | National: ReVALUE

Designed and deployed production AI systems across multiple EU Horizon Europe programs in healthcare, smart mobility, and environmental monitoring – operating within regulated, multi-country, multi-stakeholder environments.

- **Medical AI:** Engineered an NLP-driven Clinical Decision Support System (CDSS) automating MAGGIC risk scoring from unstructured clinical notes - reducing manual assessment time by ~40%, with LLM-generated patient and specialist recommendations and explanations, validated under clinical specialist supervision.
- **Neurodegenerative Disease Health:** Developed a lightweight Custom CNN (Keras/TensorFlow) for Facial Expression Recognition, matching benchmark accuracy with significantly lower model complexity than VGG-Face — reducing compute costs and enabling deployment on resource-constrained, low-cost hardware suitable for clinical settings
- **Smart Mobility:** Developed mobility analytics pipeline (Flask, PostgreSQL) with 30% faster data retrieval via SQL. Also, Delivered Random Forest transport classifier (85 % accuracy, 5 modes) adopted in municipal planning. Co - developed the [HIT application](#) backend.
- **Real Estate & Environmental:** Delivered ML pricing model ($R^2 = 0.80$, 10K+ properties) in collaboration with Tech Open House, and multi-modal sensor pipelines (LiDAR, RGB, thermal) for hazard detection.

Medical Physics and Data Analysis Intern

AHEPA University of Hospital of Thessaloniki

Thessaloniki, GR (June 2019 - Sep 2019)

- Assisted with clinical data collection, analysis, and reporting under supervision of medical physics department.

Independent Tutor & Academic Instructor

Sep 2018 – Apr 2026

- Delivered university-level Biomechanics instruction and STEM education (Physics, Mathematics, Informatics) to secondary and college-level students, including Anatolia College. Developed structured lesson materials and assessment frameworks.

RESEARCH & PROJECTS

AI & DL Group

Athens, GR (Sep. 2025 – Present)

- **AI Content & Support Automation Crew (CrewAI)** — Multi-agent orchestration system sequential & collaborative workflows (research/write/edit pipeline, customer support automation, outreach tooling). [[GitHub Repo](#)]
- **Markdown Document Preprocessor:** — Full-stack web app converting 12+ file formats to Markdown for LLM/RAG pipelines. Deployed on Hugging Face Spaces. ([MarktDown](#)).
- **Euro-Demographics Intelligence Platform:** Interactive dashboard analysing demographic trends across 30+ European countries (2000–2025). Integrated Random Forest & ARIMA forecasting models with an LLM-powered Q&A agent for real-time data exploration. Deployed on Hugging Face Spaces. ([Euro_Demographics](#))

EDUCATION

Master's in AI & Deep Learning (90 ECTS)

Athens, GR (2025 - 2026)

University of West Attica - English-taught program - Electrical & Electronics Engineering

- **Thesis:** AI Assistant — A Multimodal Conversational Agent with Realistic Avatar for Physics Education

Master's in Health Statistics and Data Analytics (90 ECTS)

Thessaloniki, GR (2022 - 2024)

Aristotle University of Thessaloniki - English-taught program - Medicine / Electrical Engineering and Informatics

- **Thesis:** Kopalidis, T., E. Samoli (2023). Effect of short-term exposure to ambient temperature on pediatric hospital admissions in Athens. <https://doi.org/10.26262/heal.auth.ir.353982>

Bachelor's in Physics (244 ECTS)

Aristotle University of Thessaloniki

Thessaloniki, GR (2020)

- **Project:** Applications of Spect γ-Camera in the study of the effect of the acquisition radius on the semiquantitative measurements futing brain straitum phantom spect imaging.

CERTIFICATIONS

ReGeneration Academy – 22nd Cycle by The Coca-Cola Company

Athens, GR (May 2026 – Jun. 2026)

- Selective employability program. Completed training in Business Readiness, Career Readiness, and AI Readiness (Presentation & Communication, Project Management, AI tools).

AI Skills Passport — EY & Microsoft

Athens, GR (2026)

- Certificate of Completion covering employability sections in AI, Sustainability, Business & Entrepreneurship, and Technology.

Microsoft Azure AI Essentials Professional Certificate

Thessaloniki, GR (2026)

NVIDIA certification

Thessaloniki, GR (2026)

- Adding New Knowledge to LLMs, Rapid Application Development with Large Language Models (LLMs)

Certification on PostgreSQL

Thessaloniki, GR (2026)

- The Complete SQL Bootcamp (Udemy): Go from Zero to Hero – Udemy in 2 months training seminar.

The AI Engineer Course 2026: Complete AI Engineer Bootcamp

Thessaloniki, GR (2026)

- Complete AI Engineer Training: Python, NLP, Transformers, LLMs, LangChain, Hugging Face, APIs (30 hours) – Udemy.

PUBLICATIONS

1. Nikiforiadis, A....., **Kopalidis, T.**, et al. (2026) Transforming Urban Mobility Behavior through Gamification and Local Incentives in Thessaloniki, Greece (accepted).

2. Kopalidis, T., et al. (2024). Advances in facial expression recognition: a survey of methods, benchmarks, models, and datasets. *Information*, 15(3), 135. DOI: [10.3390/info15030135](https://doi.org/10.3390/info15030135) ([Highly Cited Article](#) - top100 most-cited authors in information)

ADDITIONAL INFORMATION

Languages Greek (Native), English (C2), French (B2 in progress)

Interests Marathon Athlete, World Travel, Read History Books, Gym, Coding, Sudoku

Military Service Fulfilled